ASYNCHRONOUS ACTIVITY NO.2

One Molecule, Four Representations: From Molecular Formula to Bond-Line Structure

Angel Stefanie C. Gutierrez
BSChE-3B

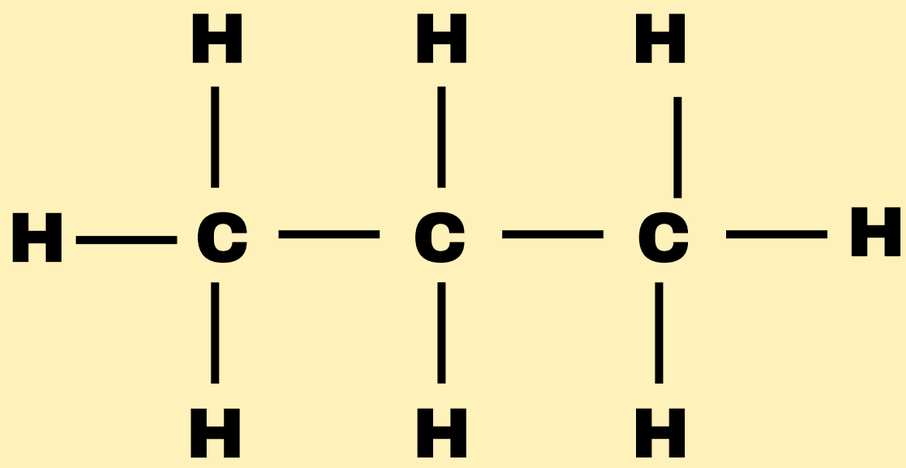
Card 01 – PROPANE

Molecular Formula : C_3H_8

1. MOLECULAR FORMULA

C_3H_8

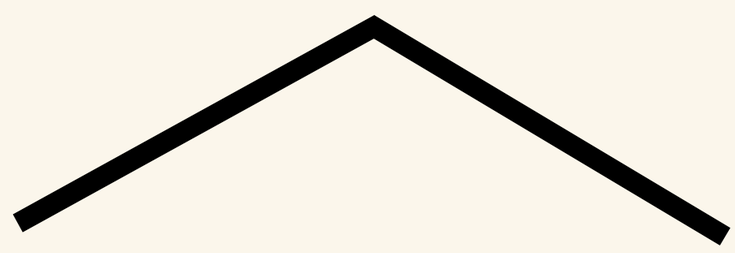
2. STRUCTURAL FORMULA



3. CONDENSED STRUCTURAL FORMULA

$CH_3CH_2CH_3$

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

- ✅ atom count verified
- ✅ valence checked
- ✅ bond order checked
- ✅ same connectivity confirmed

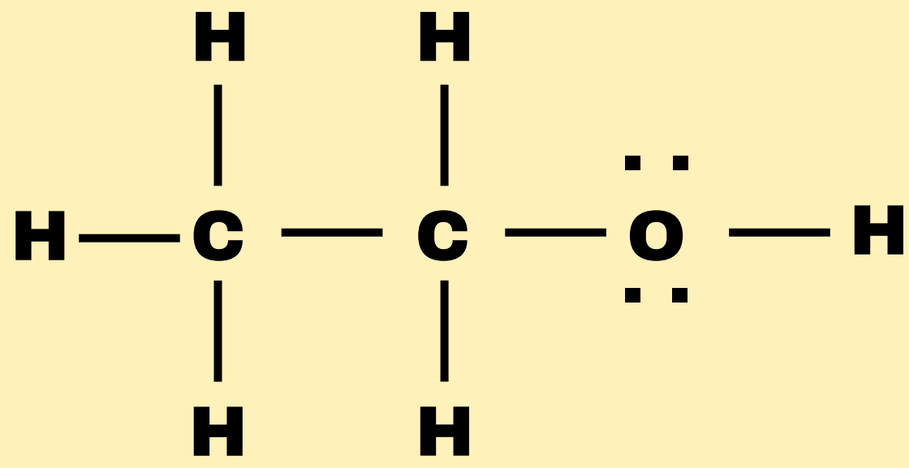
CARD 02 - ETHANOL

Molecular Formula : C_2H_6O

1. MOLECULAR FORMULA

C_2H_6O

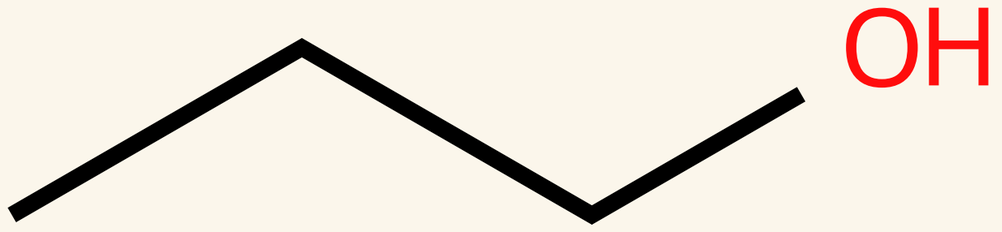
2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

CH_3CH_2OH

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

- atom count verified
- valence checked
- bond order checked
- same connectivity confirmed

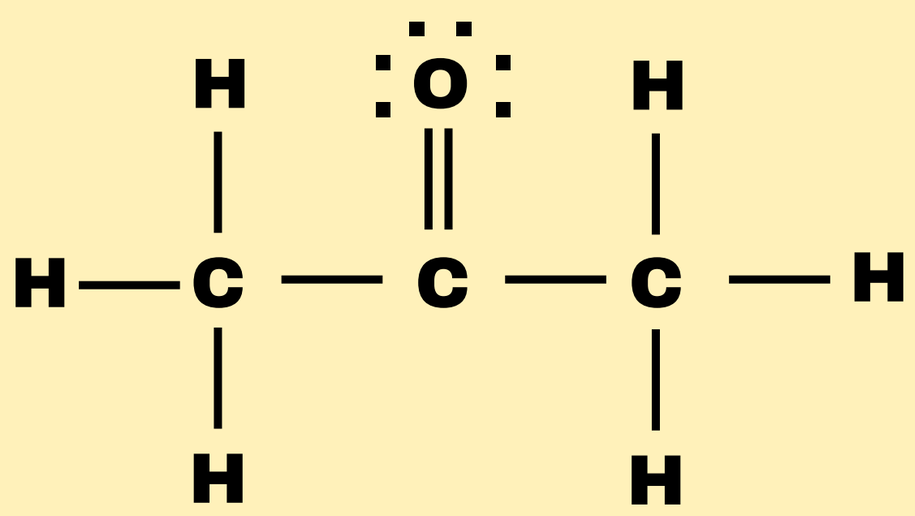
CARD 03 - ACETONE (PROPAN-2-ONE)

Molecular Formula : C_3H_6O

1. MOLECULAR FORMULA

C_3H_6O

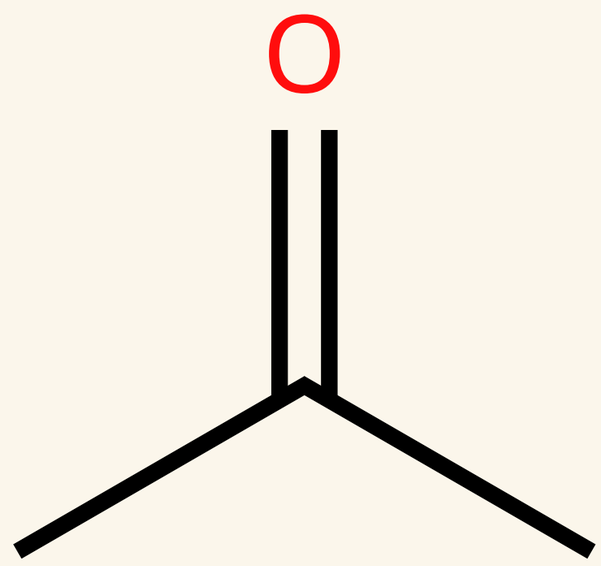
2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

CH_3COCH_3

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

- atom count verified
- valence checked
- bond order checked
- same connectivity confirmed

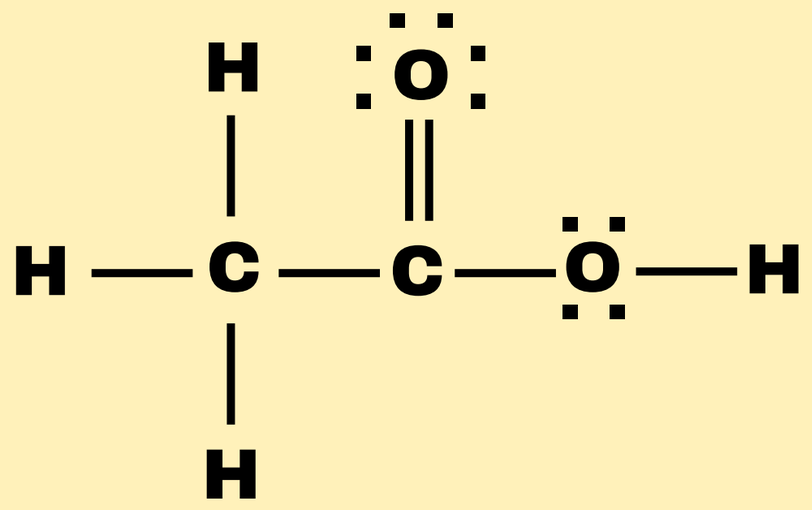
CARD 04 – ACETIC ACID (ETHANOIC ACID)

Molecular Formula : $C_2H_4O_2$

1. MOLECULAR FORMULA

$C_2H_4O_2$

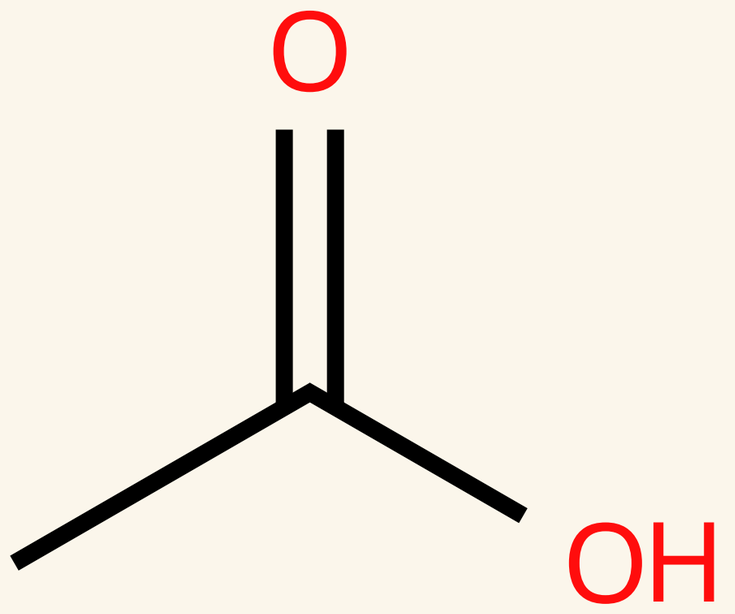
2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

CH_3COOH

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

- atom count verified
- valence checked
- bond order checked
- same connectivity confirmed

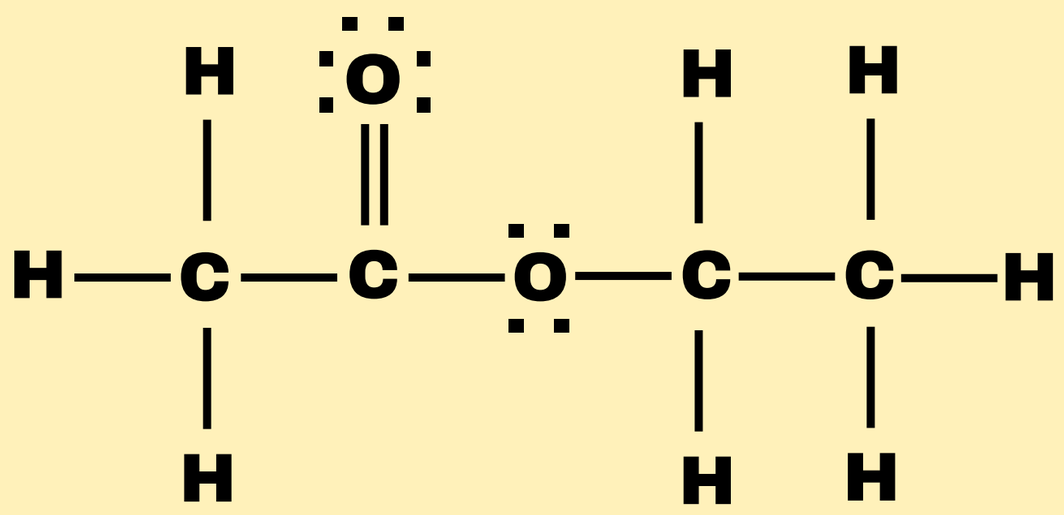
CARD 05 - ETHYL ACETATE (ETHYL ETHANOATE)

Molecular Formula : $C_4H_8O_2$

1. MOLECULAR FORMULA

$C_4H_8O_2$

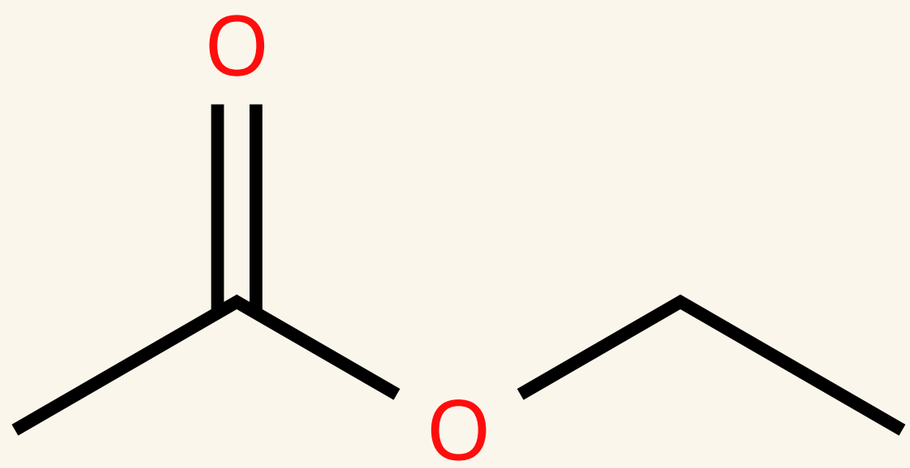
2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$CH_3COOCH_2CH_3$

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

- atom count verified
- valence checked
- bond order checked
- same connectivity confirmed

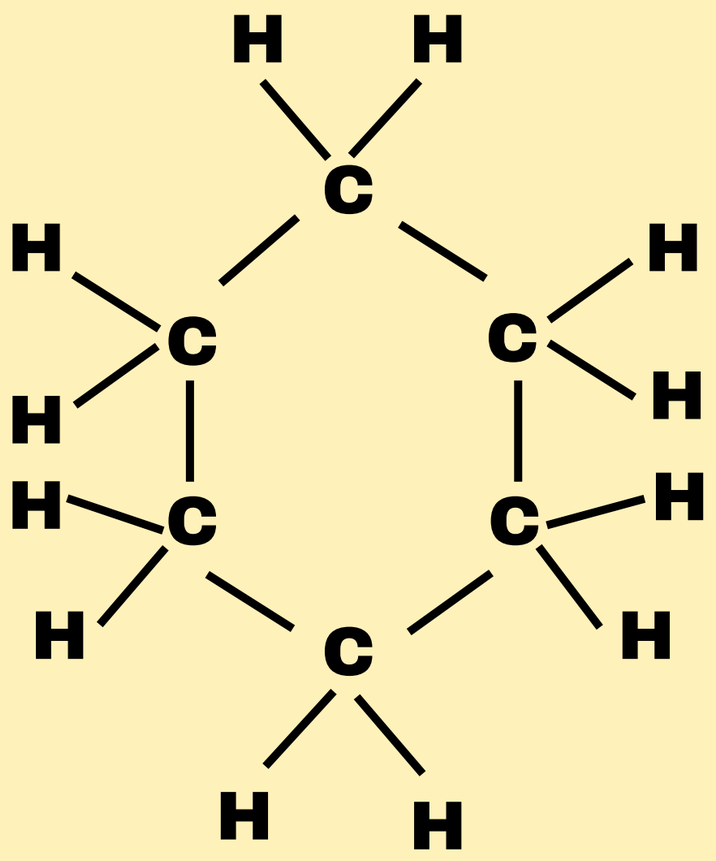
CARD 06 - CYCLOHEXANE

Molecular Formula : C_6H_{12}

1. MOLECULAR FORMULA

C_6H_{12}

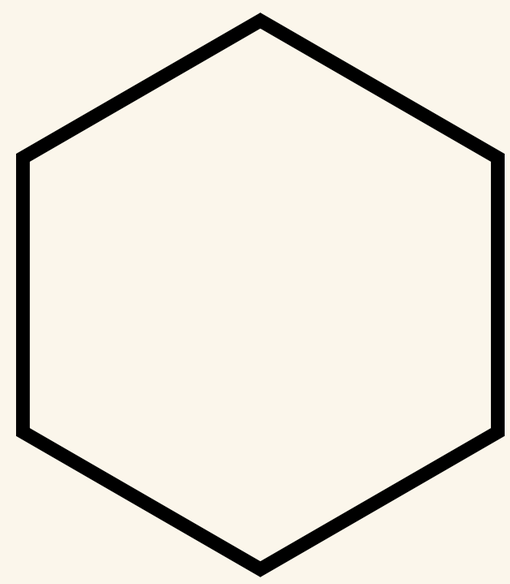
2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$(CH_2)_6$

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

- ✅ atom count verified
- ✅ valence checked
- ✅ bond order checked
- ✅ same connectivity confirmed

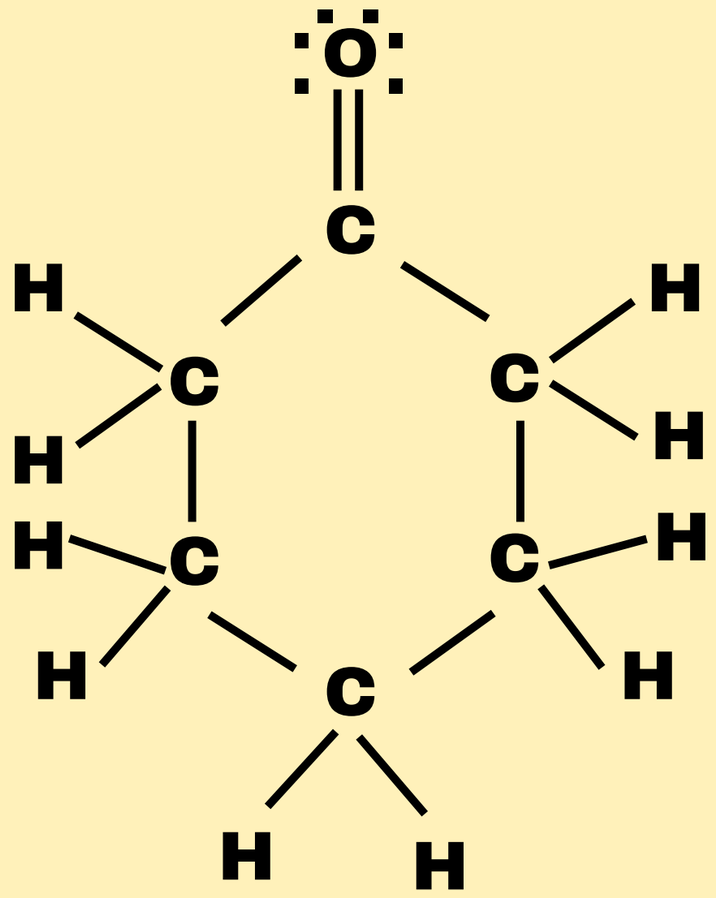
CARD 07 - CYCLOHEXANONE

Molecular Formula : $C_6H_{10}O$

1. MOLECULAR FORMULA

$C_6H_{10}O$

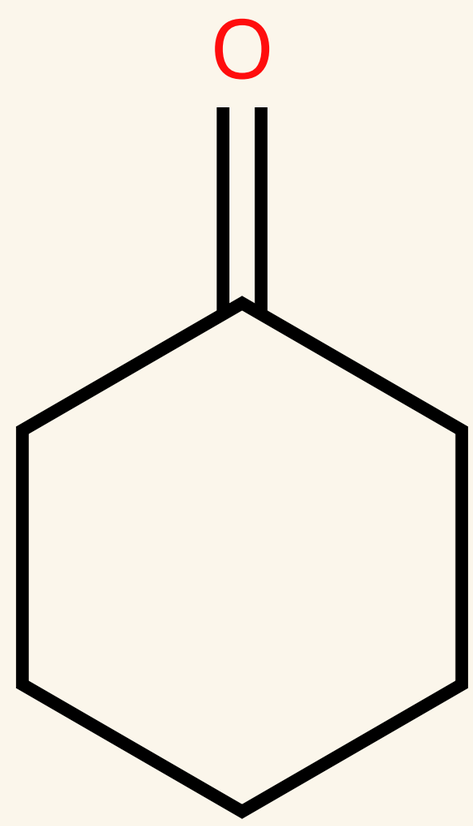
2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$(CH_2)_5CO$

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

- atom count verified
- valence checked
- bond order checked
- same connectivity confirmed

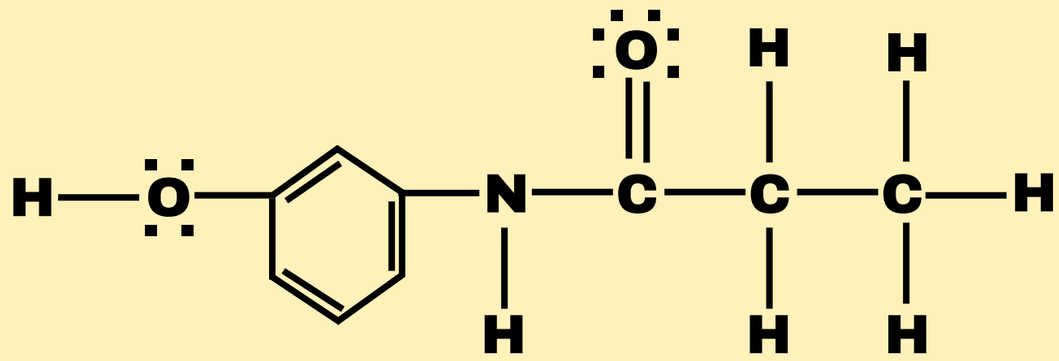
CARD 08 - ACETAMINOPHEN (PARACETAMOL)

Molecular Formula : $C_8H_9NO_2$

1. MOLECULAR FORMULA

$C_8H_9NO_2$

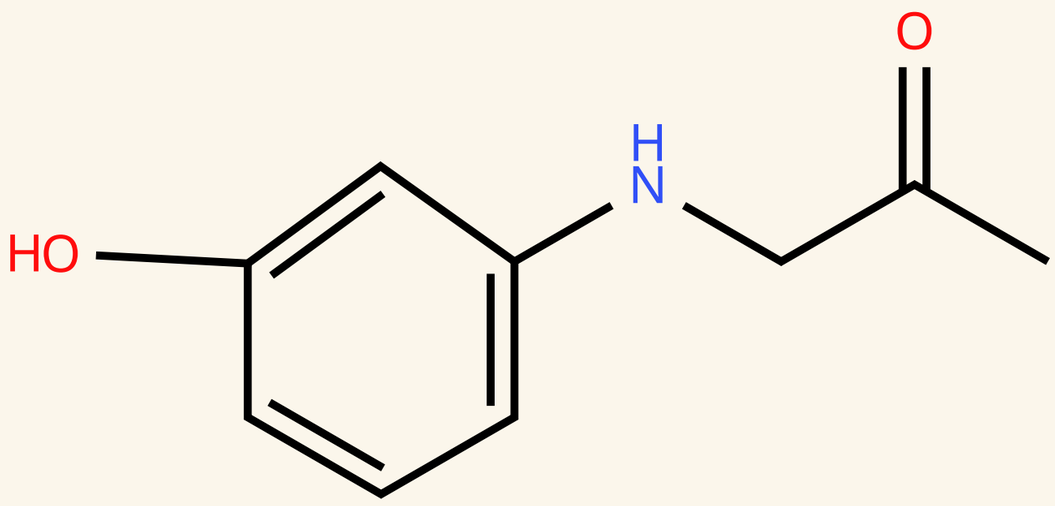
2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$HOC_6H_4NHCOCH_3$

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

- atom count verified
- valence checked
- bond order checked
- same connectivity confirmed

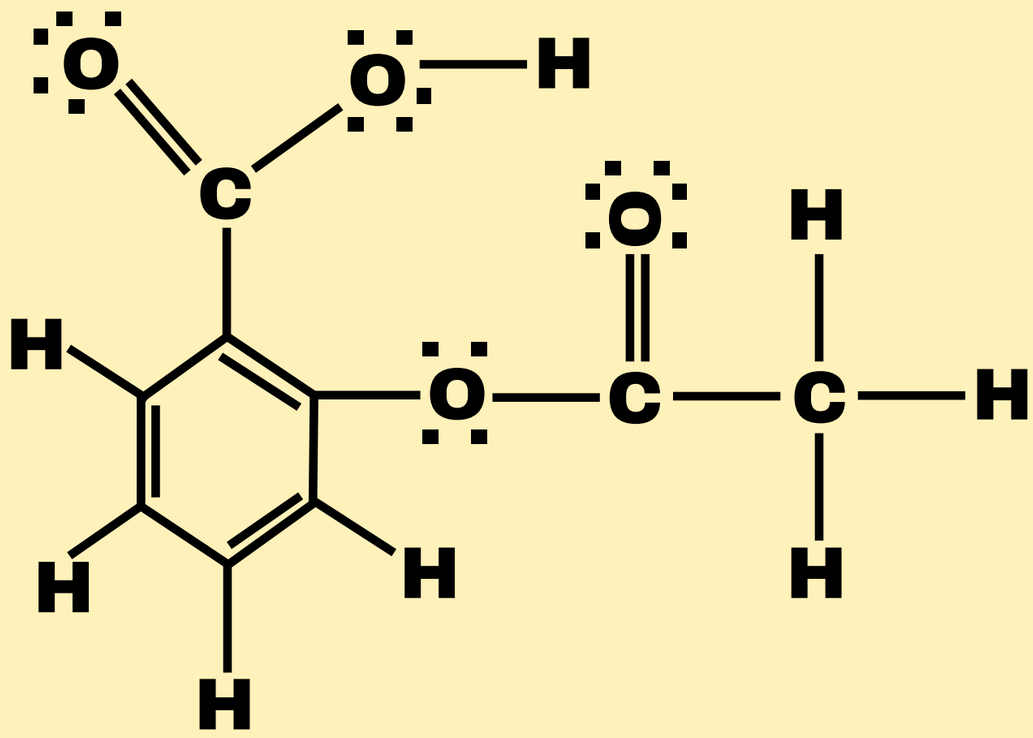
CARD 09 – ASPIRIN (ACETYLSALICYLIC ACID)

Molecular Formula : $C_9H_8O_4$

1. MOLECULAR FORMULA

$C_9H_8O_4$

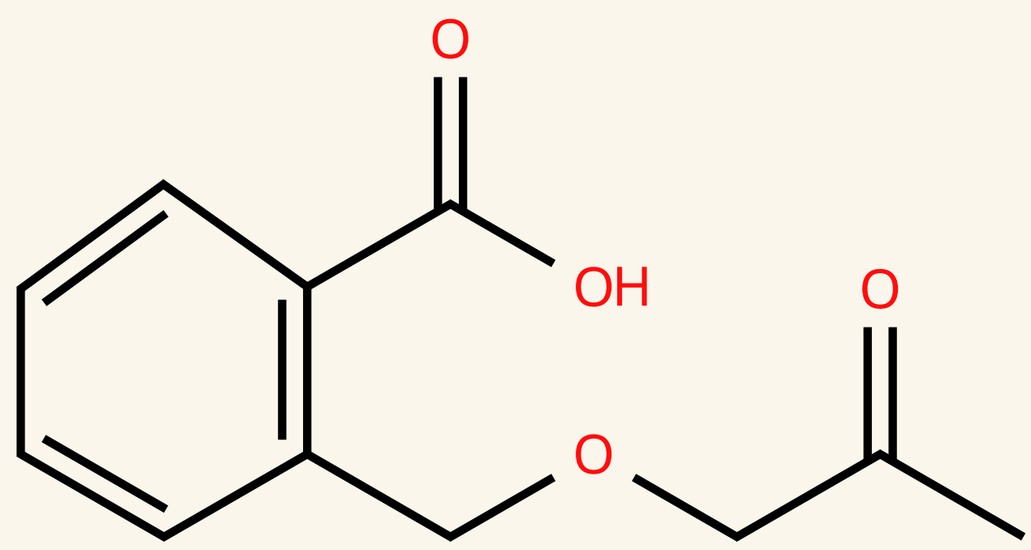
2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$CH_3COOC_6H_4COOH$

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

- atom count verified
- valence checked
- bond order checked
- same connectivity confirmed

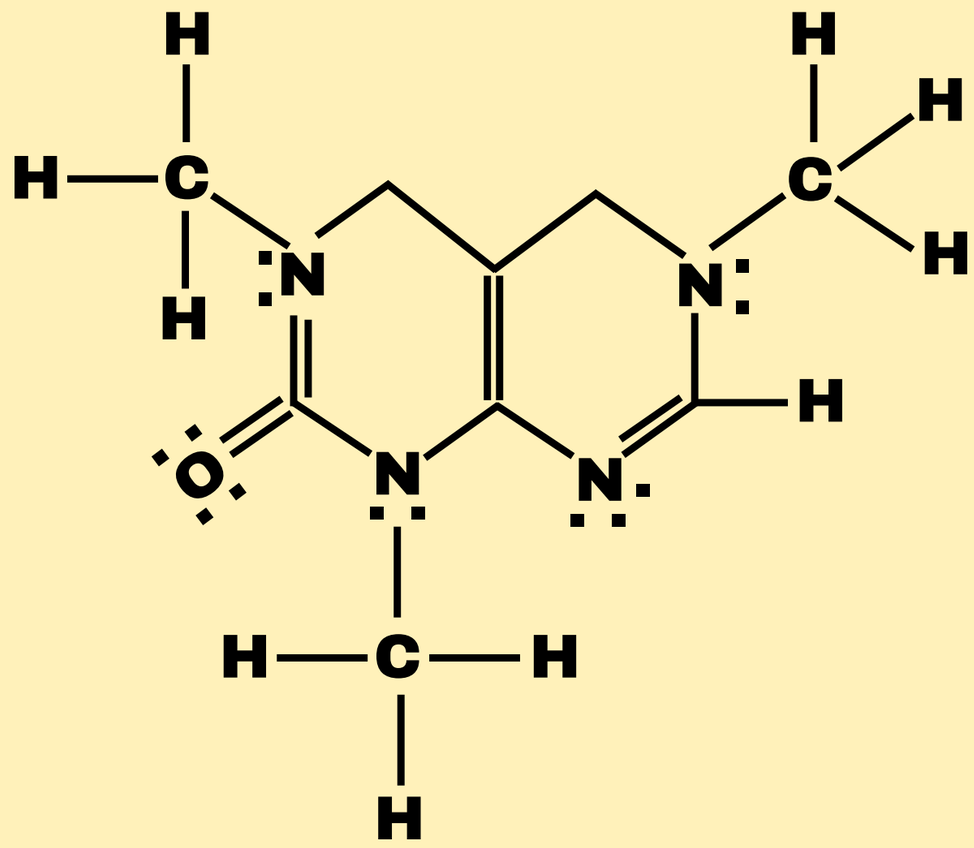
CARD 10 - CAFFEINE

Molecular Formula : $C_8H_{10}N_4O_2$

1. MOLECULAR FORMULA

$C_8H_{10}N_4O_2$

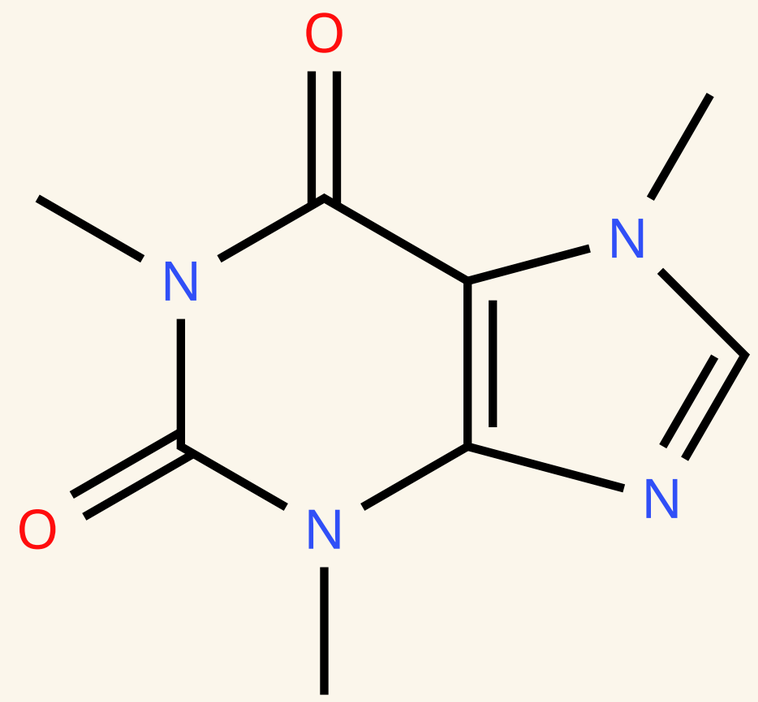
2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$CH_3N(C=O)C(CH_3)$
$=NC=NC(N(C)$
$C=O)=C1$

4. BOND-LINE FORMULA (KETCHER)



VERIFICATION STRIP

- ✅ atom count verified
- ✅ valence checked
- ✅ bond order checked
- ✅ same connectivity confirmed